



Python Analytic

Course Name	Python Analytic	
Total Course Hours	30 Hours	
Course Date	Saturday : 5-Jul, 12-Jul, 19-Jul, 26-Jul, 2-Aug, 9-Aug, 16-Aug, 23-Aug Tuesday : 19-Aug, 26-Aug	
Course Time	7:00pm-10:00pm	
Total Course Fee	HKD20000 / Applicant	
Course Venue	Class A FLAT/RM.B7/F SUN SHINE CENTRE 61-63 PORTLAND STREET , YAU MA TEI , Kowloon	
Course Objective	This course aims to introduce students to the dynamic world of data analysis and visualization, with a focus on applications in the business and finance sectors. Students will gain a comprehensive understanding of what data analytics entails and how it is applied in real-world scenarios. By the end of the course, students will be able to use industry-standard tools and acquire the necessary knowledge and skills to pursue a career as a data analyst in business or scientific research. Python, as a versatile and general-purpose programming language, serves as the backbone of this course. The course will provide hands-on experience with datasets from the finance sector, fostering students' critical thinking and problem-solving abilities. Ultimately, students will be able to perform data analysis using Python and will have a portfolio of projects demonstrating their ability to analyze and interpret complex data, preparing them for careers in finance, business analysis, and beyond.	
Course Object	Business Owner/Company Executive/Key Staff	
Teaching Language	Cantonese and English Terminology	
Module Hours	Module1 2 Hours	Module2 8 Hours
	Module3 6 Hours	Module4 10 Hours
	Module5 4 Hours	



Python Analytic

Module Details

Module1: Introduction to System Design and Python

Duration: 2 hours

- IT Industries
- Introduction to System Design
- Introduction to Data Analysis
- Introduction to SDLC
- Computer System
- Hardware and Software
- System Software and Application Software
- Integrated Development Environment
- Developer Tools

Module2: Introduction to Python

Duration: 8 hours

- Installation
- Integrated Development Environment
- Start your first Python application
- Bytecode interpreter
- Variable, String, Number, Comment,
- List, Tuple, Dictionary and Set
- If statement
- Loop(iteration)
- Function, Lambda
- Decorator
- Namespace, Variable Scope
- Mini-Project

(Financial data collecting methods, Basic data handling)

Module3: Statistic and Algebra for Data Analysis

Duration: 6 hours

- Statistics
- Population and Sample
- Parameter and Statistics
- Representativeness
- Linear Regression
- Matrix and its basic operation
- Real world applications
- Multiple Regression
- Multiple Regression Model
- Mini-Project

(Assets Correlation Analysis, Math in Financial Market)

Module4: Advanced Python

Duration: 10 hours

- Python Class & Object
- Class Inheritance
- Composition and Component
- Module and Package
- Files handling
- Exceptions
- Generator
- Mini Project – Sentiment Analysis
- Basic Time Series Analysis
- Patterns Recognition
- Autocorrelation
- ARIMA
- Rolling Statistics
- Mini-Project

(Stationary and Non-stationary Data in Finance, Standardization and Normalization of Data)

Module5: Numpy

Duration: 4 hours

- Data type, operator, comparison
- Reshape, flatten, T, concatenate, split, vsplit
- Append, insert, delete, unique
- Rounding decimal
- Random, randint, rand, uniform, ndenumerate
- Scalar, vector and matrix
- Broadcasting
- Universal Function
- Filter, Conditional Statement with np.where
- Mini-Project

(Handling Massive Financial Data)

Trainer Profile:

Honcy Lee

Introduction

Mr. Lee has achieved a high level of professionalism in finance and accounting, supported by a strong educational foundation and extensive teaching experience. He holds a BA in Accounting & Finance and a PgD in Financial Analytics and Algo Trading with merits and distinction award. As an experienced educator, Mr. Lee has taught over 600 hours of Python Data Analytics at several institutes in Hong Kong. He excels at communicating complex concepts and empowering students with the analytical skills needed to thrive in data-driven environments. His expertise in both the theoretical and practical aspects of finance, accounting, and data analysis makes him a valuable asset in any educational or professional setting.

Mr. Lee is also the founder and director of a company that provides information technology and financial data analytics services. He has served as a managing director in a private tech company and has over a decade of experience working with several multinational financial institutions.

Expertise

- Data Analysis Tools: Python/SQL/Pandas/Numpy/Polars/Numba/SciPy
- Machine Learning: SKlearn/XGBOOST/TensorFlow/Keras
- Data Integration: Web Scraping/Beautifulsoup/Selenium/Scrapy/Session management
- Cloud Services: Google Colab/AWS cloud
- Version Control and Collaboration: Git/GitHub
- Automation: Scheduled Task in all Mac/Linux/Windows
- Web development: Django/HTML/CSS/Flowbite/API
- Corporate Financial Report/Accounting Report analysis
- Startup kickstart project